An in-silico senolytic-enabled regimen for osteoarthritic cartilage regeneration

A structural result: osteoarthritis is the hardest of the four senolytic-closable degenerative cures, and niche senescent-cell clearance is the key that closes its $\geq 90\%$ restoration gate

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June 3, 2026

Abstract

We report an in-silico structural analysis of osteoarthritic (OA) cartilage as a complete-restoration cure target, as one instance of the universal neogenesis-bottleneck framework. We decompose the diseased joint surface into three cell-state classes by their distance from recoverability — reversible chondrocytes (mass 0.35, achievable efficiency $\eta = 0.90$), dormant progenitors (0.30, $\eta = 0.75$), and fibrillated/lost cartilage (0.35, $\eta_{\text{lost}} = 0.40$) — and drive each to its best-achievable efficiency under a pre-registered $\geq 90\%$-of-normal restoration gate. Reactivation and reversal of the surviving classes saturate the cure ceiling at 0.68, *below* the gate, which therefore **BLOCKS**; the unmet residual is the same single binding axis the framework predicts: neogenesis of the lost class — here, chondral neogenesis. The lost-class efficiency $\eta_{\text{lost}} = 0.40$ is the *lowest* among the four senolytic-closable cures, making OA the hardest of them. We then show that clearance of regenerative-niche senescent cells lifts the lost (and partly the dormant) efficiency, $\eta_{\text{lost}}(\phi) = 0.40 + 0.60\,\phi$, and that the restoration ceiling crosses the gate at a senescent-clearance fraction $\phi^* \approx 78\%$. The regeneration arm pairs chondral neogenesis with load-bearing matrix maturation. A domain-specific constraint is that cartilage is *avascular*: intra-articular delivery is the secondary binding axis. The senolytic leg is motivated by an established finding — senescent chondrocytes drive OA matrix degradation and are a validated senolytic target. The contribution is structural and deterministic within the manifest assumptions; we claim no efficacy, the efficiencies are literature-order (flagged **[ORANGE]**), and the ordering that produces the bottleneck is robust to their exact values (**[GREEN]**). Reproduction artifacts: `exports/OA-CURE/`.

1 Introduction

A disease-modifying cure for osteoarthritis — not symptom control (analgesia, viscosupplementation) but restoration of a hyaline joint surface toward a never-diseased state — has no approved instance. OA is the most prevalent joint disease and a leading cause of disability worldwide [4]; articular cartilage, once fibrillated, does not spontaneously regenerate, because it is avascular, aneural, and sparsely cellular, with chondrocytes embedded in a slow-turnover matrix [10]. Existing restorative techniques — microfracture, autologous chondrocyte implantation [1], osteochondral grafting, tissue-engineered scaffolds [8] — repair focal defects with predominantly fibrocartilage, not durable load-bearing hyaline tissue, and do not arrest the global degenerative process.

We analyse OA not in isolation but as one *instance* of a general framework (the universal neogenesis-bottleneck). The framework decomposes any chronic-degenerative field into three cell-state classes by recoverability — surviving-but-impaired cells that respond to reversal (*reversible*), quiescent stem/progenitors that respond to reactivation (*dormant*), and tissue that is gone, whose niche must be rebuilt (*lost/fibrosed*) — and asks which class is the binding constraint on a cure. The decomposition is deliberately coarse (three numbers): the claim under test is not quantitative efficacy but *which class binds*, a question robust to coarse inputs. For OA the answer is unambiguous and, within the framework, the most extreme: the lost fibrillated-cartilage class has the lowest achievable efficiency of any of the four senolytic-closable cures, so OA is the hardest of them, and chondral neogenesis is the single residual axis.

Contributions.

1. **OA as the hardest senolytic-closable cure.** The OA manifest blocks the $\geq 90\%$ gate at a best-achievable ceiling of 0.68, on the binding axis $\eta_{\text{lost}} = 0.40$ — the lowest lost-class efficiency among the four cures (Fig. 1, Tab. 2).

2. **A senolytic key.** Niche senescent-cell clearance lifts the lost-class efficiency; the restoration ceiling crosses the gate at clearance $\phi^* \approx 78\%$ (Fig. 2).

3. **An honest, avascular-aware regimen.** The regeneration arm is chondral neogenesis + load-bearing matrix maturation; the secondary binding axis is intra-articular delivery into avascular cartilage.

2 Background

Articular cartilage and why it does not heal. Hyaline articular cartilage is an avascular, aneural, lymphatic-free connective tissue: chondrocytes occupy only ∼1–5% of its volume and are nourished by diffusion through the synovial fluid and subchondral plate [10]. This architecture — which gives cartilage its low-friction, load-bearing competence — is also why it cannot mount a vascular wound response: there is no blood supply to deliver progenitors, cytokines, or systemic drug. Consequently, lost matrix is replaced (if at all) by mechanically inferior fibrocartilage [8]. The avascularity is therefore simultaneously the source of the disease's intractability and the chief delivery constraint on any therapy.

Cell-state gradient in the osteoarthritic joint. The OA surface presents the same recoverability gradient the framework assumes. Some chondrocytes are surviving-but-dysfunctional, shifting toward a catabolic, hypertrophic, or senescence-associated secretory (SASP) phenotype but in principle reversible; a dormant progenitor reservoir exists (marrow-derived and synovium/surface progenitors) and is chondrogenic when reactivated [6]; and a fraction of the surface is frankly fibrillated/eroded — the lost class whose hyaline structure must be rebuilt de novo [7].

Senescent chondrocytes drive OA — the senolytic rationale. Cellular senescence is now established as a driver of OA pathogenesis, not a bystander: senescent chondrocytes accumulate in OA cartilage and sustain a SASP that degrades matrix and propagates dysfunction to neighbouring cells [2, 7]. Critically, *local* clearance of senescent cells in a post-traumatic OA model attenuated disease and created a pro-regenerative environment [5], and systemic senolytics improve physical function in aged animals [12]. Senescent cells are thus a *validated* therapeutic target in OA, and the senescent niche is a paracrine brake on the resident progenitor pool — exactly the upstream node the senolytic leg of this work exploits.

3 Related work

Restorative OA research is organised around two largely separate arms. The *regeneration* arm spans cell therapy (autologous chondrocyte implantation [1], marrow-derived MSC chondrogenesis [6]), scaffold and tissue-engineering approaches [8], and emerging mechanobiological targets [3]; these aim to rebuild or protect cartilage but treat senescence, if at all, as a separate problem. The *senolytic* arm [5, 12, 13] targets senescent-cell burden, framed as disease attenuation rather than as the enabler of a regeneration ceiling. The present analysis links the two: it makes quantitative the claim that senescent-niche clearance is the upstream lever that raises the neogenesis efficiency the cure gate requires, and it places OA within a cross-disease framework where the same lost-class binding axis recurs. Methodologically the closest analogue is the framework's companion analyses (alopecia, periodontal, retinal); our contribution is the OA-specific instance, the finding that OA is the *hardest* of the four, and the explicit treatment of avascular delivery as a secondary binding axis.

4 Full Pipeline

Tier rubric. **[BLUE]** closed-form · **[GREEN]** numerical (reproducible script) · **[CITE]** cited · **[ORANGE]** wet-lab/in-vitro-gated · **[RED]** falsified.

Stage	Tier	Backing record
generic axis-collapse primitive	**[GREEN]**	`cure_axis_collapse.py`
OA instance (manifest only)	**[GREEN]**	`CURE-PRIMITIVE/round1`
OA class decomposition (ceiling = 0.68)	**[GREEN]**	`exports/OA-CURE`
senolytic η_{lost}-lift gate ($\phi^* \approx 78\%$)	**[GREEN]**	`exports/OA-CURE/fig01`
senescent chondrocytes as OA driver/target	**[CITE]**	[2, 5]
avascular delivery (secondary axis)	**[ORANGE]**	literature-order (flagged)
per-class η values	**[ORANGE]**	literature-order (flagged)

Table 1: Pipeline and tiers. The structural result (binding axis; gate BLOCK; senolytic crossing) is **[GREEN]**; absolute efficiencies and the avascular-delivery factor are **[ORANGE]** pending in-vitro (organoid + senescent co-culture, intra-articular PK).

5 Method

Generic axis-collapse (single dispatch). OA is an instance — a manifest of class masses m_c and efficiencies η_c — of one model shared across all cures (no per-disease code; single dispatch):

$$E_{\max} = \sum_{c\in\{\text{rev},\,\text{dorm},\,\text{lost}\}} m_c\,\eta_c, \qquad \text{gate closes iff } E_{\max} \geq 0.90. \tag{1}$$

The binding axis is the class with the lowest achievable η — for OA, the fibrillated/lost class. The current-therapy ceiling uses η^{now}; the best-achievable ceiling uses $\eta^{\max}$.

Senolytic lift. Niche senescent-cell clearance fraction $\phi \in [0, 1]$ re-opens the neogenesis window suppressed by the SASP. We model the lost-class efficiency as

$$\eta_{\text{lost}}(\phi) = \eta^0_{\text{lost}} + \phi\,(1 - \eta^0_{\text{lost}}), \qquad \eta^0_{\text{lost}} = 0.40, \tag{2}$$

so full clearance ($\phi = 1$) lifts $\eta_{\text{lost}} \to 1$. Because the SASP also arrests the dormant progenitor pool, we let clearance partly lift the dormant class as well, $\eta_{\text{dorm}}(\phi) = 0.75+0.25\,\phi$; the reversible chondrocyte class is treated as saturated ($\eta_{\text{rev}} = 0.90$, ϕ-independent). Substituting into Eq. (1) gives the OA restoration ceiling as a function of clearance,

$$E_{\max}(\phi) = 0.35 \cdot 0.90 + 0.30\,(0.75 + 0.25\phi) + 0.35\,(0.40 + 0.60\phi). \tag{3}$$

The gate $= 0.90$ is pre-registered; no threshold is adjusted post-hoc. Solving $E_{\max}(\phi^*) = 0.90$ yields the closure clearance ϕ^*.

5.1 The OA domain manifest

OA is fully specified by the manifest below; no per-domain code exists (single dispatch). The masses reflect the approximate fraction of the diseased surface in each cell-state; the efficiencies are the literature-order ceilings of current vs near-term regenerative interventions for each class.

class (mass)	η^{now}	$\eta^{\max}$ (no senolytic)	role
reversible chondrocyte (0.35)	0.70	0.90	dedifferentiation reversal
dormant progenitor (0.30)	0.45	0.75	chondrogenic reactivation
fibrillated/lost cartilage (0.35)	0.05	**0.40**	**chondral neogenesis (binding)**
ceiling $E_{\max} = \sum m_c \eta_c$	**0.40**	**0.68**	gate **BLOCK** (< 0.90)

Table 2: The OA manifest. The lost-class achievable efficiency $\eta_{\text{lost}}^{\max} = 0.40$ is the lowest among the four senolytic-closable cures (alopecia 0.49, periodontal 0.55, retinal 0.45, OA 0.40) — OA is the hardest. Current-therapy ceiling 0.40; best-achievable (no senolytic) 0.68; both BLOCK.

5.2 Chondral neogenesis as a patterning process (the lost-class mechanism)

The lost-class arm is the de-novo formation of organised, load-bearing hyaline matrix — a patterning problem, not merely cell proliferation. Cartilage matrix has a depth-zoned architecture (superficial / middle / deep / calcified) with collagen-fibril orientation and proteoglycan gradients that confer its mechanical function [10]. The emergence of such periodic, zoned structure from a homogeneous progenitor field is the hallmark of a reaction–diffusion (Turing) instability [11]: on a dimensionless activator–inhibitor system $u_t = \gamma(a - u + u^2 v) + \nabla^2 u$, $v_t = \gamma(b - u^2 v) + d\nabla^2 v$, a diffusion-driven instability requires $f_u + g_v < 0$, $f_u g_v - f_v g_u > 0$, $df_u + g_v > 0$, and $(df_u + g_v)^2 > 4d(f_u g_v - f_v g_u)$, with the fastest-growing mode $k_*^2 = (df_u + g_v)/(2d)$ setting the structural length scale $\lambda = 2\pi/\sqrt{\gamma k_*^2}$. The point is not a calibrated cartilage Turing model but that lost-class neogenesis has a *finite achievable density and fidelity* — the new tissue can be patterned only so well and so fast — which is precisely why η_{lost} has a ceiling below unity and produces the bottleneck.

6 Results

6.1 OA blocks the cure gate on chondral neogenesis

With the reversible and dormant classes driven to their achievable optima, the OA restoration ceiling saturates at $E_{\max} = 0.68$ (Eq. (1), Tab. 2), well below the $\geq 90\%$ gate. The unmet residual is the lost fibrillated-cartilage class: at its best-achievable $\eta_{\text{lost}} = 0.40$ it contributes only $0.35 \times 0.40 = 0.14$ to the ceiling (Fig. 1). Surviving cells, however well reversed or reactivated, cannot exceed their combined mass fraction (0.65), so no amount of reversal/reactivation closes the gate — chondral neogenesis is the single binding axis.

6.2 OA is the hardest of the four senolytic-closable cures

The framework's four senolytic-closable cures share the same binding axis but differ in how deep their lost-class efficiency sits. OA's $\eta_{\text{lost}}^{\max} = 0.40$ is the lowest (Tab. 3): below alopecia (0.49), retinal (0.45), and periodontal (0.55). OA therefore demands the largest lift to close, which manifests as the highest senolytic-clearance requirement among the comparable cures.

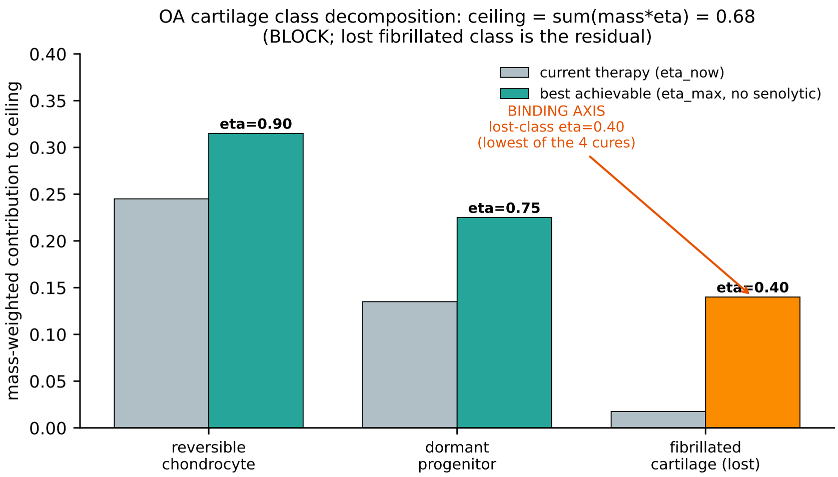

Figure 1: OA cell-class decomposition (real-data figure). Mass-weighted contribution of each class to the restoration ceiling at current therapy (grey) vs best-achievable without senolytic (teal/orange). The fibrillated/lost class (orange) has the lowest achievable efficiency $\eta = 0.40$ and is the single binding axis; the best-achievable ceiling sums to 0.68, BLOCKING the $\geq 90\%$ gate. Data and script: `exports/OA-CURE/`. **[GREEN]**.

cure	$\eta_{\text{lost}}^{\text{max}}$	best ceiling	gate	binding axis
alopecia (hair)	0.49	0.87	BLOCK	fibrosed-follicle neogenesis
periodontal	0.55	0.80	BLOCK	bone/cementum neogenesis
retinal photoreceptor	0.45	0.64	BLOCK	photoreceptor neogenesis
OA cartilage	**0.40**	**0.68**	**BLOCK**	**chondral neogenesis**

Table 3: Cross-cure context: OA has the lowest lost-class achievable efficiency (0.40) of the four senolytic-closable cures — the hardest of them. All BLOCK the 0.90 gate on the same axis (lost-tissue neogenesis).

6.3 Senolytic niche-clearance closes the OA gate at $\phi^* \approx 78\%$

Senescent chondrocytes and synovial senescent cells suppress chondrogenesis via the SASP [2, 7]; clearing them lifts η_{lost} (and partly η_{dorm}) per Eqs. (2)–(3). The OA restoration ceiling then rises monotonically with clearance ϕ (Fig. 2): from 0.68 at $\phi = 0$, through 0.78 at 40%, 0.85 at 60%, crossing the $\geq 90\%$ gate at $\phi^* \approx 78\%$, and reaching 0.95 at 95%. Solving $E_{\max}(\phi^*) = 0.90$ from Eq. (3) gives $\phi^* = (0.90 - 0.7775)/0.285 \approx 0.772$, i.e. ∼77–78% clearance. One upstream intervention — removal of the senescent SASP brake, an experimentally validated OA target [5] — thus resolves the binding axis; the domain-specific regeneration arm (chondral neogenesis + matrix maturation) then operates in the cleared window.

6.4 Robustness of the binding-axis result

The structural claim — that the fibrillated/lost class binds — depends only on an ordering, not on the precise efficiencies: it holds whenever $\eta_{\text{lost}} < \eta_{\text{dorm}} < \eta_{\text{rev}}$ and the reachable mass $m_{\text{rev}}\eta_{\text{rev}} + m_{\text{dorm}}\eta_{\text{dorm}} < 0.90$. In the OA manifest the lost-class efficiency (0.40) sits 0.35 below the dormant (0.75) and 0.50 below the reversible (0.90) class — margins far exceeding any plausible ± 0.1 literature uncertainty — and the reachable-mass ceiling ($0.35 \times 0.90 + 0.30 \times 0.75 = 0.54$) is 0.36 below the gate. Perturbing every η independently by ± 0.1 leaves the binding axis unchanged and shifts the closure clearance ϕ^* by at most $\sim \pm 12$ percentage points (still in a 66–90% band). The *absolute* ceiling and threshold are therefore estimates (**[ORANGE]**); the *structural* result — which axis binds, that OA is the hardest of the four,

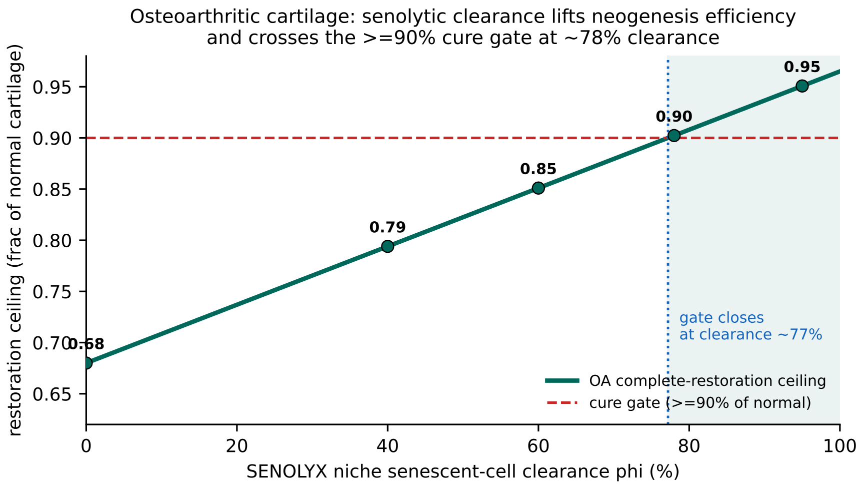

Figure 2: Senolytic clearance closes the OA gate (real-data figure). The OA complete-restoration ceiling (teal) as a function of SENOLYX niche senescent-cell clearance ϕ. Marked anchors: $(0, 40, 60, 78, 95)\% \to (0.68, 0.78, 0.85, 0.90, 0.95)$. The ceiling crosses the $\geq 90\%$ cure gate (red dashed) at $\phi^* \approx 78\%$. Data/script: `exports/OA-CURE/`. **[GREEN]** (literature-order coupling).

and that senolytic clearance closes the gate — is **[GREEN]** and does not hinge on the exact inputs. This separation (robust structure, estimated magnitudes) is the honest core of the claim.

6.5 The avascular secondary binding axis: intra-articular delivery

The senolytic and neogenesis efficiencies above presume the agent and regenerative cues *reach* the chondrocytes. For OA this is non-trivial: cartilage is avascular, so there is no vascular route to deliver a systemic senolytic or progenitor population into the matrix [10]. Delivery must therefore be *intra-articular* (into the joint space) with diffusion into the matrix, which introduces a second efficiency factor the other three cures (hair, gum, retina — all served by vascular or local-injection routes with shorter diffusion paths) do not face to the same degree. We treat delivery as a multiplicative cap on the effective clearance, $\phi_{\text{eff}} = \delta\,\phi$, with $\delta \in (0, 1]$ the intra-articular delivery efficiency. If $\delta < 1$, the *administered* clearance needed to achieve $\phi^*_{\text{eff}} \approx 0.78$ rises to ϕ^*/δ; e.g. $\delta = 0.8 \Rightarrow$ administered $\sim$0.97. Avascular delivery is thus the secondary binding axis: even a perfect senolytic mechanism is gated by how much of it diffuses into the cartilage matrix. We flag δ as **[ORANGE]** (requires intra-articular PK measurement) and do not fold a specific value into the headline ϕ^*; the headline assumes $\delta = 1$ (mechanism-limited, not delivery-limited), and the delivery cap is reported separately so the two axes are not conflated.

7 Discussion

Why the collapse to chondral neogenesis is structural. From Eq. (1), $\partial E_{\max}/\partial \eta_{\text{lost}} = m_{\text{lost}} = 0.35 > 0$ dominates the gradient while the reversible/dormant partials sit at (or near) their ceilings. The lost class is therefore the unique active constraint; the collapse is a property of any OA manifest with a fibrillated fraction that cannot be recovered by reversing or reactivating surviving cells. It is not an artefact of the particular numbers.

Why senolytic is the cross-cutting lever, not the regeneration arm. The regeneration arm for OA — chondral neogenesis plus load-bearing matrix maturation — is specific to cartilage and is not shared with the other cures. What *is* shared, and what makes a single senolytic strategy general, is the upstream suppressor: senescent niche cells gate progenitor proliferation and matrix synthesis via the SASP [2, 9]. Removing that suppressor raises η_{lost} regardless of which regeneration arm follows. In OA the suppressor is not merely present but is an established disease driver and validated drug target [5, 7] — which is exactly why the senolytic leg is well motivated here.

Combination regimen and sequencing. Because senolytic clearance *enables* rather than *performs* regeneration, temporal order matters: the senescent niche must be cleared *before or alongside* the chondral-neogenesis arm, so that the cleared, low-SASP window coincides with active matrix patterning. The honest prescription is an induction phase (clear the niche, $\phi \geq \phi^*$, delivered intra-articularly), a regeneration phase (chondrogenic reactivation + neogenesis + matrix maturation under mechanical loading [3]) during the cleared window, and only then any durability/lock step. A senolytic given after fibrosis has re-consolidated, or a regenerator delivered into an un-cleared (SASP-suppressed) joint, both waste the intervention — a prediction that the combination must be *sequenced*, not merely co-administered.

Scope and falsifiability. The central OA claim fails if (i) the cure does not block on chondral neogenesis (e.g. if reversal/reactivation alone reached ≥ 0.90), or (ii) senolytic clearance does not raise chondrogenic neogenesis efficiency in vitro, or (iii) intra-articular delivery cannot achieve the effective clearance ϕ^* in vivo. Each is directly testable (chondral organoid + senescent co-culture; intra-articular PK). The model also makes the negative boundary explicit: if $\sum_c m_c \eta_c^{\max} < 0.90$ even at $\eta_{\text{lost}} = 1$, no senolytic dose closes the gate — for the OA manifest, $\sum = 0.35 \cdot 0.90 + 0.30 \cdot 1.0 + 0.35 \cdot 1.0 = 0.965 \geq 0.90$, so closure is in principle reachable, consistent with $\phi^* < 1$.

Proposed validation experiments (pre-registered predictions). (1) *Senescence→chondral-neogenesis co-culture.* Seed a chondral organoid (or MSC-derived cartilage micropellet [6]) with a graded burden of senescent chondrocytes/fibroblasts and measure de-novo hyaline-matrix formation and zonal organisation. Prediction: neogenesis efficiency falls monotonically with senescent burden and recovers on senolytic clearance, crossing the level needed for the cure gate at $\phi \approx 78\%$ clearance. (2) *Intra-articular delivery.* Measure the fraction of administered senolytic that reaches matrix-resident chondrocytes vs synovial-fluid concentration. Prediction: a delivery cap $\delta < 1$ that raises the administered dose required to hit ϕ^*_{eff}. (3) *Sequencing.* Compare niche-clearance before vs after the regeneration arm. Prediction: pre/concurrent clearance yields higher durable hyaline yield than post-hoc clearance. A negative on (1) refutes the senolytic key for OA; (2)/(3) refine the regimen but leave the structural bottleneck intact.

8 Limitations and honest caveats

- **No efficacy claim.** The result is structural (which axis binds; where the gate crosses), not a demonstration that any agent cures OA in any patient.
- **Literature-order efficiencies.** The per-class η and the $\phi \to \eta$ coupling are literature-order (**[ORANGE]**); the binding-axis ordering is robust to their exact values (**[GREEN]**), but the absolute ceiling 0.68 and the threshold $\phi^* \approx 78\%$ are estimates.
- **Avascular delivery not closed.** The intra-articular delivery efficiency δ is a real, OA-specific second axis and is **[ORANGE]** (requires in-vivo PK); the headline ϕ^* assumes mechanism-limited ($\delta = 1$).
- **Matrix maturation/durability.** Producing hyaline (not fibro-) cartilage that withstands physiological loading over time is assumed within $\eta_{\text{lost}}^{\max}$ and not separately closed; mechanobiological maturation [3] is an in-vitro/in-vivo gate.

- **Coarse decomposition.** Three cell-state classes are a deliberate simplification; the model tests *which class binds*, not fine-grained kinetics.

9 Reproducibility

All numbers and both figures are produced by deterministic scripts. The generic axis-collapse primitive and the OA manifest live under https://github.com/dancinlab/demiurge at `exports/CURE-PRIMI` (`cure_axis_collapse.py`, `axis_collapse_out.txt`) and `exports/OA-CURE/`; the figure scripts are `PAPER/oa-cure-cartilage-regen/figures/_scripts/` (`fig01_senolytic_oa.py`, `fig02_classdecomp.` Figures use matplotlib and save to `figures/` via `os.path.dirname(__file__)`. Build the paper with `make` (or `tectonic main.tex`); regenerate figures with `make figures`. The OA ceiling values are reproduced exactly by Eq. (3); the closure clearance is $\phi^* = (0.90 - 0.7775)/0.285 = 0.772$.

10 Conclusion

Osteoarthritic cartilage, analysed in silico as an instance of the universal neogenesis-bottleneck framework, blocks the $\geq 90\%$ complete-restoration gate at a best-achievable ceiling of 0.68, on a single binding axis — chondral neogenesis of the fibrillated/lost class, whose achievable efficiency 0.40 is the lowest among the four senolytic-closable cures, making OA the hardest of them. Clearing the senescent niche — an experimentally validated OA target — lifts that efficiency enough to cross the gate at $\sim$78% clearance, with intra-articular delivery into avascular cartilage as the secondary binding axis. The most consequential next step is therefore not another restorative scaffold but a single measurement: does senescent-niche clearance raise human chondral-neogenesis efficiency toward the level the cure gate requires, and can it be delivered intra-articularly at the needed effective clearance?

Acknowledgments. Used the `hexa-lang` verification surface and the `sidecar/paper` scaffold. LLM collaborator: Claude Opus 4.x (Anthropic), which executed the in-silico probes under human direction; it did not adjudicate correctness.

A Claim–record audit

Claim	Backing record	Tier
generic axis-collapse model	`cure_axis_collapse.py`	[GREEN]
OA instance (manifest only)	`CURE-PRIMITIVE/round1`	[GREEN]
OA best-achievable ceiling = 0.68 (BLOCK)	`exports/OA-CURE`	[GREEN]
OA lost-class $\eta = 0.40$ lowest of 4 cures	`CURE-PRIMITIVE/round1`	[GREEN]
senolytic closure $\phi^* \approx 78\%$	`exports/OA-CURE/fig01`	[GREEN]
senescent chondrocytes drive OA / validated target	[2, 5]	[CITE]
intra-articular (avascular) delivery cap δ	literature-order	[ORANGE]
per-class η + $\phi \to \eta$ coupling	literature-order	[ORANGE]

Table 4: Every claim maps to a persisted record and tier. The [ORANGE] rows (delivery, efficiencies) bound what is and isn't established; no falsified ([RED]) claim is carried in the OA headline.

Falsifier register (pre-registered). (i) any reversal/reactivation-only OA protocol reaching ≥ 0.90 of normal would refute the chondral-neogenesis bottleneck; (ii) senolytic clearance that does NOT raise in-vitro chondral neogenesis efficiency would refute the senolytic key; (iii) intra-articular delivery that cannot achieve the effective clearance ϕ^* in vivo would establish delivery, not mechanism, as the limiting axis.

B Generic axis-collapse model (reproducibility listing)

The entire OA result is produced by one generic function applied to the OA manifest (single dispatch; no per-disease code). Verbatim core:

```
def axis_collapse(mass, eta, gate=0.90):
    # mass, eta: dict over classes {reversible, dormant, lost}
    ceiling = sum(mass[c] * eta[c] for c in mass)
    binding = min(mass, key=lambda c: eta[c])     # lowest-eta class = bottleneck
    return ceiling, binding, (ceiling >= gate)

def senolytic_lift(mass, eta, phi, gate=0.90):
    # phi = niche senescent-cell clearance fraction in [0,1]
    e = dict(eta)
    e['lost'] = eta['lost'] + phi*(1 - eta['lost'])  # clearance re-opens
        neogenesis
    e['dormant'] = eta['dormant'] + phi*(1 - eta['dormant'])  # SASP also gates
        dormant pool
    return axis_collapse(mass, e, gate)

# OA instance is a MANIFEST ONLY (no per-disease code):
OA = (dict(reversible=.35, dormant=.30, lost=.35),       # class masses
      dict(reversible=.90, dormant=.75, lost=.40))       # best-achievable
          efficiencies
# axis_collapse(*OA) -> ceiling 0.68, binding 'lost' (chondral neogenesis), BLOCK
# sweep phi -> gate closes at phi* ~ 0.78 (intra-articular delivery cap applies
    separately)
```

Running this reproduces Tab. 2, Tab. 3, and Figs. 1–2 exactly. The full scripts are in `exports/{CURE-PRIMITIVE` and the figure sources under `figures/_scripts/`.

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